21 The diagram shows a rectangular wall.



Frank is going to cover the wall with rectangular tiles.
Each tile is 60 cm by 30 cm.

$\frac{3}{5}$ of the tiles will be white.
Some of the tiles will be green.
The rest of the tiles will be blue.

The ratio of the number of green tiles to the number of blue tiles will be 1 : 3

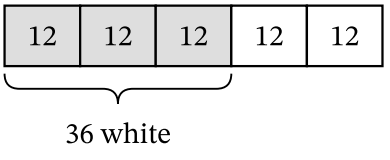
(a) Assuming there are no gaps between the tiles, how many tiles of each colour will Frank need?

600 ÷ 60 = 10 tiles across; 180 ÷ 30 = 6 high.

Total = 10 × 6 = 60 tiles.

2 marks total
Correct tile count
60 tiles

60 tiles

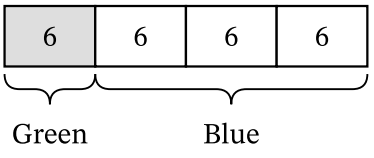


One fifth = 60 ÷ 5 = 12.

White = 3 × 12 = 36.

1 mark
3/5 of 60

24 left



60 − 36 = 24 left.

One part = 24 ÷ 4 = 6.

white tiles 36

green tiles 6

blue tiles 18

1 mark
Share remaining 24 in 1:3

1 mark
White 36; green 6; blue 18

(5)

Frank is told that he should leave gaps between the tiles.

(b) If Frank leaves gaps between the tiles, how could this affect the number of tiles he needs?

Fewer tiles may be needed: the gaps take up some area.

1 mark
Fewer tiles may be needed

(1)

(Total for Question 21 is 6 marks)